Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Biometrics [BME 4056]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Evaluate the relative merits of biometric access control systems versus traditional card-and-password systems for securing controlled drug storage rooms in a hospital setting. Consider factors such as security, convenience, and cost-effectiveness. (3)

1B) Analyze the impact of poor feature extraction on the overall performance of a biometric system. Provide two specific consequences and explain their effects. (3)

1C) Analyze the potential trade-offs between user acceptability and system security in the design of a biometric system for mobile devices. (4)

2A) Explain how ridge endings and bifurcations are used in fingerprint recognition systems. Provide an example of how these features are extracted from a fingerprint image. (3)

2B) Compare the effectiveness of using geometric-based features versus texture-based features in facial recognition. Discuss the advantages and disadvantages of each approach. (4)

2C) Explain how the unique patterns in the iris are used in iris recognition systems. Provide an example of how these patterns are extracted and utilized for identity verification. (3)

3A) Given the ROC of two systems A and B as shown in Figure 1. Analyze the characteristics and compare the performances of the systems. (3)

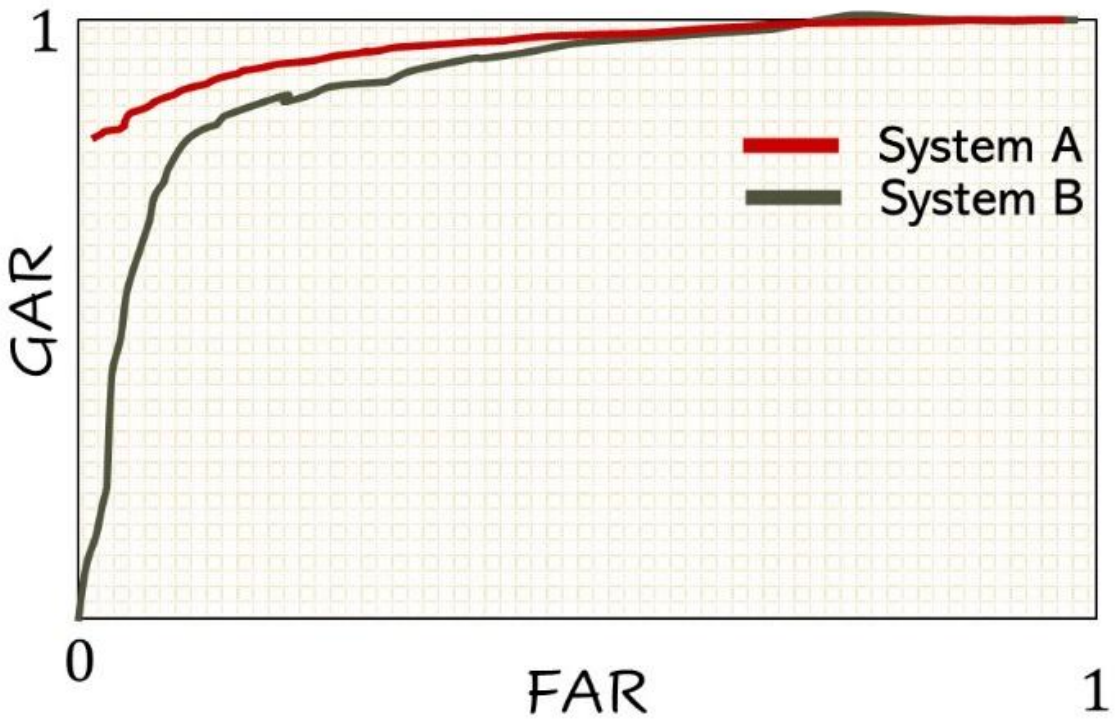


Figure 1

3B) Given two biometric systems, System A and System B, with the following error rates:

•System A: FAR = 0.02%, FRR = 0.05%

•System B: FAR = 0.01%, FRR = 0.1% (4)

If each system processes 20,000 verification attempts, calculate the total number of errors (false acceptances + false rejections) for each system and determine which system is more reliable.

3C) For the biometric system depicted in Figure 2, determine the FAR and FRR regions corresponding to the chosen threshold. Discuss the implications of these regions on the system's overall performance in terms of its trade-off between security and convenience. (3)

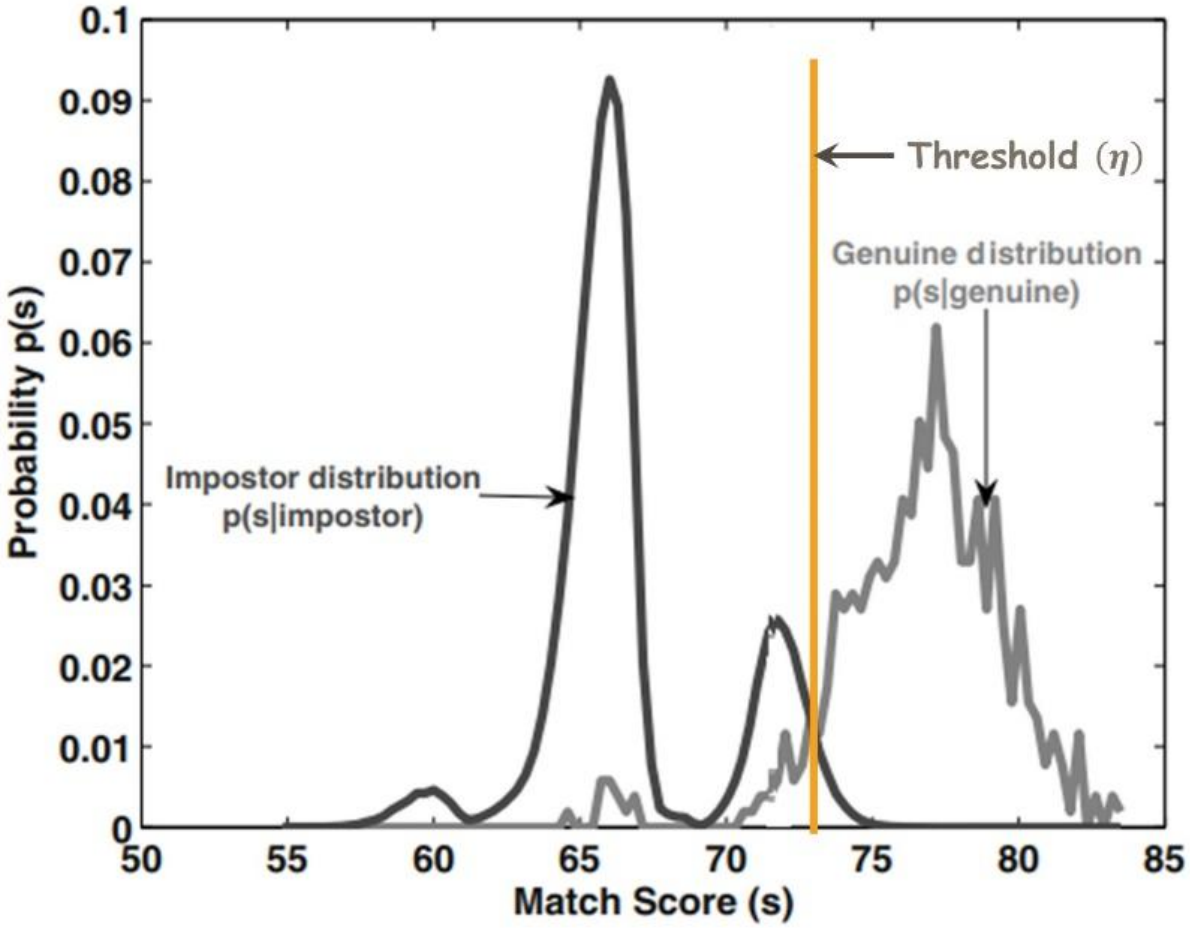


Figure 2

4A)	Justify the choice of Tanh normalization for score-level fusion in this biometric system, considering its ability to handle outliers, preserve relative differences, and improve overall system performance.	(4)
4B)	Analyze the potential shortcomings of using Min-Max normalization for score-level fusion in this biometric system?	(4)
4C)	Outline the challenges in designing the multibiometric system.	(2)
5A)	Analyze the potential consequences of false identifications in biometric systems, focusing on their impact on individual privacy and integrity.	(4)
5B)	Evaluate the role of Privacy by Design (PbD) in mitigating privacy risks associated with biometric systems and discuss the specific privacy-enhancing technologies that can be implemented.	(3)
5C)	Compare and contrast the ethical implications of first-generation biometrics (fingerprinting, facial recognition) and second-generation biometrics (behavioral, emotional tracking) in terms of privacy, consent, and potential for misuse.	(3)

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